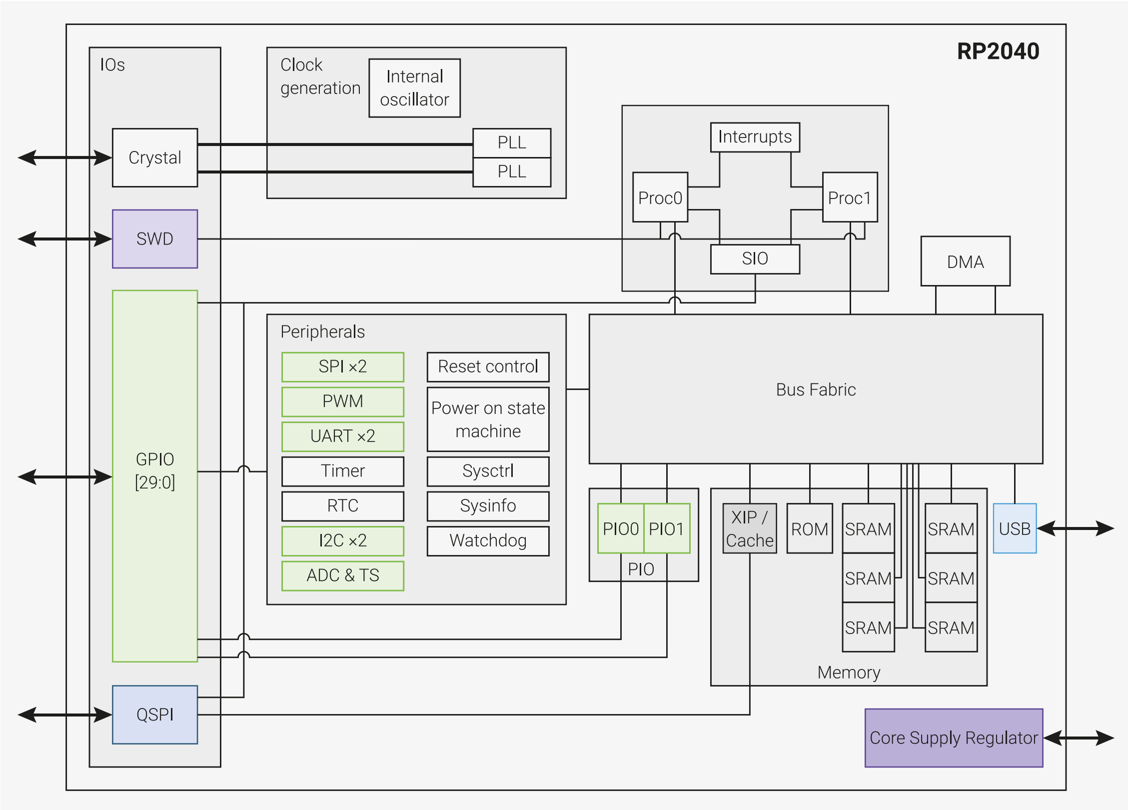


Figure 2. A system overview of the RP2040 chip

Code may be executed directly from external memory through a dedicated SPI, DSPI or QSPI interface. A small cache improves performance for typical applications.

Debug is available via the SWD interface.

Internal SRAM can contain code or data. It is addressed as a single 264 kB region, but physically partitioned into 6 banks to allow simultaneous parallel access from different masters.

DMA bus masters are available to offload repetitive data transfer tasks from the processors.

GPIO pins can be driven directly, or from a variety of dedicated logic functions.

Dedicated hardware for fixed functions such as SPI, I2C, UART.

Flexible configurable PIO controllers can be used to provide a wide variety of IO functions.

A USB controller with embedded PHY can be used to provide FS/LS Host or Device connectivity under software control.

Four ADC inputs which are shared with GPIO pins.

Two PLLs to provide a fixed 48MHz clock for USB or ADC, and a flexible system clock up to 133MHz.

An internal Voltage Regulator to supply the core voltage so the end product only needs supply the IO voltage.

1.4. Pinout Reference

This section provides a quick reference for pinout and pin functions. Full details, including electrical specifications and package drawings, can be found in Chapter 5.

1.4.1. Pin Locations

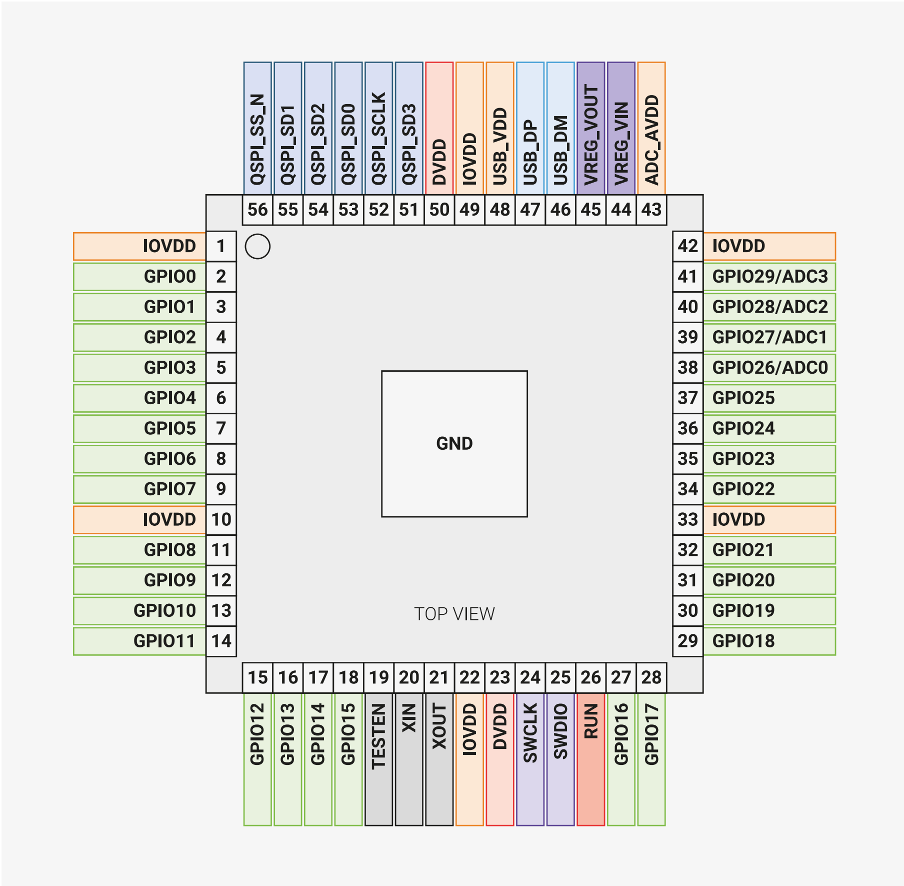


Figure 3. RP2040 Pinout for QFN-56 7×7mm (reduced ePad size)

1.4.2. Pin Descriptions

Table 1. The function of each pin is briefly described here. Full electrical specifications can be found in Chapter 5.

Name	Description
GPIOx	General-purpose digital input and output. RP2040 can connect one of a number of internal peripherals to each GPIO, or control GPIOs directly from software.
GPIOx/ADCy	General-purpose digital input and output, with analogue-to-digital converter function. The RP2040 ADC has an analogue multiplexer which can select any one of these pins, and sample the voltage.
QSPIx	Interface to a SPI, Dual-SPI or Quad-SPI flash device, with execute-in-place support. These pins can also be used as software-controlled GPIOs, if they are not required for flash access.
USB_DM and USB_DP	USB controller, supporting Full Speed device and Full/Low Speed host. A 27Ω series termination resistor is required on each pin, but bus pull-ups and pull-downs are provided internally.
XIN and XOUT	Connect a crystal to RP2040's crystal oscillator. XIN can also be used as a single-ended CMOS clock input, with XOUT disconnected. The USB bootloader requires a 12MHz crystal or 12MHz clock input. For recommended crystals, see Crystal Oscillator (Section 2.16).
RUN	Global asynchronous reset pin. Reset when driven low, run when driven high. If no external reset is required, this pin can be tied directly to IOVDD.
SWCLK and SWDIO	Access to the internal Serial Wire Debug multi-drop bus. Provides debug access to both processors, and can be used to download code.
TESTEN	Factory test mode pin. Tie to GND.
GND	Single external ground connection, bonded to a number of internal ground pads on the RP2040 die.
IOVDD	Power supply for digital GPIOs, nominal voltage 1.8V to 3.3V

Name	Description
USB_VDD	Power supply for internal USB Full Speed PHY, nominal voltage 3.3V
ADC_AVDD	Power supply for analogue-to-digital converter, nominal voltage 3.3V
VREG_VIN	Power input for the internal core voltage regulator, nominal voltage 1.8V to 3.3V
VREG_VOUT	Power output for the internal core voltage regulator, nominal voltage 1.1V, 100mA max current
DVDD	Digital core power supply, nominal voltage 1.1V. Can be connected to VREG_VOUT, or to some other board-level power supply.

1.4.3. GPIO Functions

Each individual GPIO pin can be connected to an internal peripheral via the GPIO functions defined below. Some internal peripheral connections appear in multiple places to allow some system level flexibility. SIO, PIO0 and PIO1 can connect to all GPIO pins and are controlled by software (or software controlled state machines) so can be used to implement many functions.

Table 2. General Purpose Input/Output (GPIO) Bank 0 Functions

	Function								
GPIO	F1	F2	F3	F4	F5	F6	F7	F8	F9
0	SPI0 RX	UART0 TX	I2C0 SDA	PWM0 A	SIO	PIO0	PIO1		USB OVCUR DET
1	SPI0 CSn	UART0 RX	I2C0 SCL	PWM0 B	SIO	PIO0	PIO1		USB VBUS DET
2	SPI0 SCK	UART0 CTS	I2C1 SDA	PWM1 A	SIO	PIO0	PIO1		USB VBUS EN
3	SPI0 TX	UART0 RTS	I2C1 SCL	PWM1 B	SIO	PIO0	PIO1		USB OVCUR DET
4	SPI0 RX	UART1 TX	I2C0 SDA	PWM2 A	SIO	PIO0	PIO1		USB VBUS DET
5	SPI0 CSn	UART1 RX	I2C0 SCL	PWM2 B	SIO	PIO0	PIO1		USB VBUS EN
6	SPI0 SCK	UART1 CTS	I2C1 SDA	PWM3 A	SIO	PIO0	PIO1		USB OVCUR DET
7	SPI0 TX	UART1 RTS	I2C1 SCL	PWM3 B	SIO	PIO0	PIO1		USB VBUS DET
8	SPI1 RX	UART1 TX	I2C0 SDA	PWM4 A	SIO	PIO0	PIO1		USB VBUS EN
9	SPI1 CSn	UART1 RX	I2C0 SCL	PWM4 B	SIO	PIO0	PIO1		USB OVCUR DET
10	SPI1 SCK	UART1 CTS	I2C1 SDA	PWM5 A	SIO	PIO0	PIO1		USB VBUS DET
11	SPI1 TX	UART1 RTS	I2C1 SCL	PWM5 B	SIO	PIO0	PIO1		USB VBUS EN
12	SPI1 RX	UART0 TX	I2C0 SDA	PWM6 A	SIO	PIO0	PIO1		USB OVCUR DET
13	SPI1 CSn	UART0 RX	I2C0 SCL	PWM6 B	SIO	PIO0	PIO1		USB VBUS DET
14	SPI1 SCK	UART0 CTS	I2C1 SDA	PWM7 A	SIO	PIO0	PIO1		USB VBUS EN
15	SPI1 TX	UART0 RTS	I2C1 SCL	PWM7 B	SIO	PIO0	PIO1		USB OVCUR DET
16	SPI0 RX	UART0 TX	I2C0 SDA	PWM0 A	SIO	PIO0	PIO1		USB VBUS DET
17	SPI0 CSn	UART0 RX	I2C0 SCL	PWM0 B	SIO	PIO0	PIO1		USB VBUS EN
18	SPI0 SCK	UART0 CTS	I2C1 SDA	PWM1 A	SIO	PIO0	PIO1		USB OVCUR DET
19	SPI0 TX	UART0 RTS	I2C1 SCL	PWM1 B	SIO	PIO0	PIO1		USB VBUS DET
20	SPI0 RX	UART1 TX	I2C0 SDA	PWM2 A	SIO	PIO0	PIO1	CLOCK GPIN0	USB VBUS EN
21	SPI0 CSn	UART1 RX	I2C0 SCL	PWM2 B	SIO	PIO0	PIO1	CLOCK GPOUT0	USB OVCUR DET

	Function								
22	SPI0 SCK	UART1 CTS	I2C1 SDA	PWM3 A	SIO	PIO0	PIO1	CLOCK GPIN1	USB VBUS DET
23	SPI0 TX	UART1 RTS	I2C1 SCL	PWM3 B	SIO	PIO0	PIO1	CLOCK GPOUT1	USB VBUS EN
24	SPI1 RX	UART1 TX	I2C0 SDA	PWM4 A	SIO	PIO0	PIO1	CLOCK GPOUT2	USB OVCUR DET
25	SPI1 CSn	UART1 RX	I2C0 SCL	PWM4 B	SIO	PIO0	PIO1	CLOCK GPOUT3	USB VBUS DET
26	SPI1 SCK	UART1 CTS	I2C1 SDA	PWM5 A	SIO	PIO0	PIO1		USB VBUS EN
27	SPI1 TX	UART1 RTS	I2C1 SCL	PWM5 B	SIO	PIO0	PIO1		USB OVCUR DET
28	SPI1 RX	UART0 TX	I2C0 SDA	PWM6 A	SIO	PIO0	PIO1		USB VBUS DET
29	SPI1 CSn	UART0 RX	I2C0 SCL	PWM6 B	SIO	PIO0	PIO1		USB VBUS EN

Table 3. GPIO bank 0 function descriptions

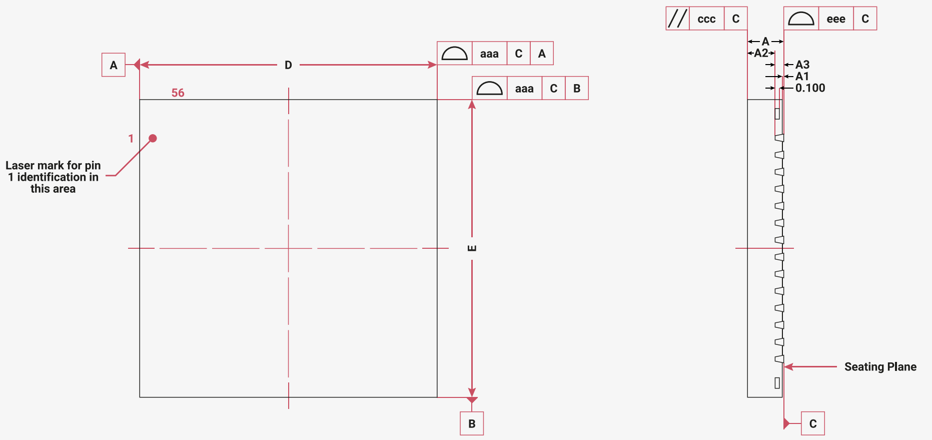
Function Name	Description
SPIx	Connect one of the internal PL022 SPI peripherals to GPIO
UARTx	Connect one of the internal PL011 UART peripherals to GPIO
I2Cx	Connect one of the internal DW I2C peripherals to GPIO
PWMx A/B	Connect a PWM slice to GPIO. There are eight PWM slices, each with two output channels (A/B). The B pin can also be used as an input, for frequency and duty cycle measurement.
SIO	Software control of GPIO, from the single-cycle IO (SIO) block. The SIO function (`F5`) must be selected for the processors to *drive* a GPIO, but the input is always connected, so software can check the state of GPIOs at any time.
PIOx	Connect one of the programmable IO blocks (PIO) to GPIO. PIO can implement a *wide* variety of interfaces, and has its own internal pin mapping hardware, allowing flexible placement of digital interfaces on bank 0 GPIOs. The PIO function (`F6`, `F7`) must be selected for PIO to *drive* a GPIO, but the input is always connected, so the PIOs can always see the state of all pins.
CLOCK GPINx	General purpose clock inputs. Can be routed to a number of internal clock domains on RP2040, e.g. to provide a 1Hz clock for the RTC, or can be connected to an internal frequency counter.
CLOCK GPOUTx	General purpose clock outputs. Can drive a number of internal clocks (including PLL outputs) onto GPIOs, with optional integer divide.
USB OVCUR DET/VBUS DET/VBUS EN	USB power control signals to/from the internal USB controller

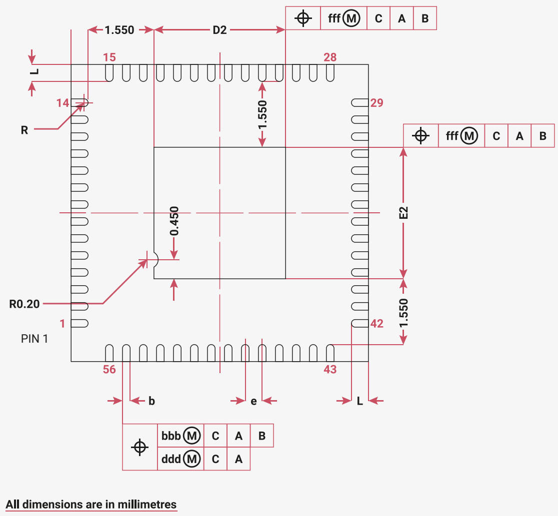
Chapter 5. Electrical and Mechanical

Physical and electrical details of the RP2040 chip.

5.1. Package

Figure 166. Top down view (left, top) and side view (right, top), along with bottom view (left, bottom) of the RP2040 QFN-56 package





Symbol	Millimetre			Inch		
	Min.	Nom.	Max.	Min.	Nom.	Max.
A	-	-	0.900	-	-	0.035
A1	0.000	-	0.050	0.000	-	0.002
A2	-	0.650	0.700	-	0.026	0.028
A3	0.203 REF			0.008 REF		
b	0.130	0.180	0.230	0.005	0.007	0.009
D	7 BSC			0.276 BSC		
D2	3.00	3.100	3.200	0.118	0.122	0.126
E	7 BSC			0.276 BSC		
E2	3.00	3.100	3.200	0.118	0.122	0.126
L	0.300	0.400	0.500	0.012	0.016	0.020
e	0.400 BSC			0.016 BSC		
R	0.065	-	-	0.003	-	-
Tolerances of form and position						
aaa	0.100			0.004		
bbb	0.070			0.003		
ccc	0.100			0.004		
ddd	0.050			0.002		
eee	0.080			0.003		
fff	0.100			0.004		

NOTE

There is no standard size for the central GND pad (or ePad) with QFNs. However, the one on RP2040 is smaller than most. This means that standard 0.4mm QFN-56 footprints provided with CAD tools may need adjusting. This gives the opportunity to route between the central pad and the ones on the periphery, which can help with maintaining power and ground integrity on cheaper PCBs. See Minimal Design Example for an example.

NOTE

Leads have a matte Tin (Sn) finish. Annealing is done post-plating, baking at 150°C for 1 hour. Minimum thickness for lead plating is 8 micromns, and the intermediate layer material is CuFe2P (roughened Copper (Cu)).

5.1.1. Thermal characteristics

The thermal characteristics of the package are shown in Table 611.

Table 611. Thermal data for the RP2040 QFN 56 package.

θ_{JA} (°C/W)	ψ_{JT} (°C/W)	ψ_{JB} (°C/W)	T_J (°C)	T_T (°C)	θ_{JC} (°C/W)	θ_{JB} (°C/W)
48.00	0.80	29.20	42.00	41.8	19.01	29.03

5.1.2. Recommended PCB Footprint

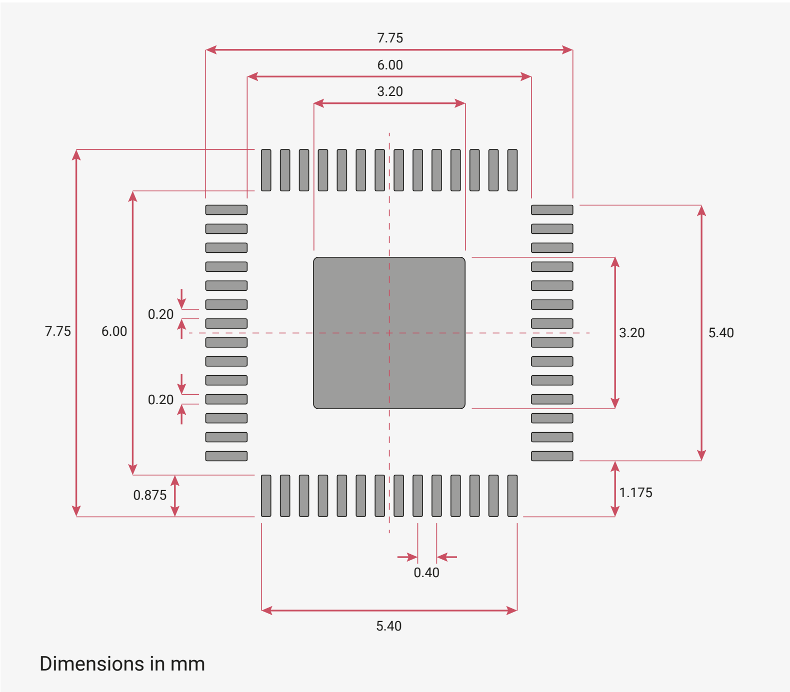


Figure 167. Recommended PCB Footprint for the RP2040 QFN-56 package

5.1.3. Package markings

The RP2040 7×7 mm QFN-56 package is marked as seen in Figure 168, with specifications as shown in Table 612. Coordinate origin is bottom-left of the package.

Figure 168. Package marking format

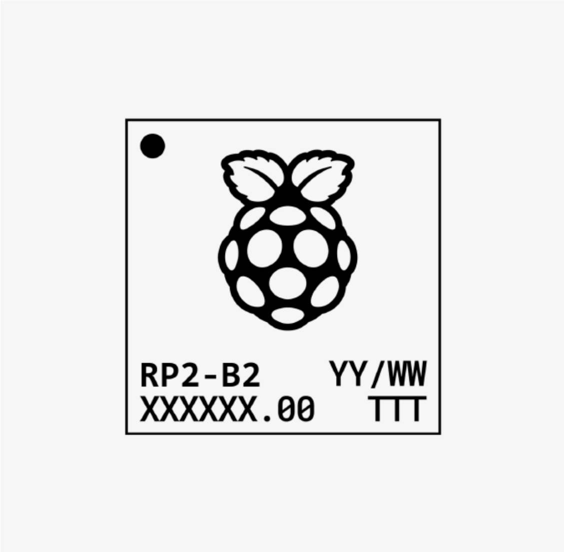


Table 612. Marking requirements and dimensions

Line	Step	Item	Coord. X	Coord. Y	Char. Height	Char. Width	Char. Space
1	1	Pin 1 Dot	0.5	6	0.5	0.5	
2	1	Logo	3.5	2.395	3.83	3.05	
3	1	RP2-B2	0.555	1.585	0.61	0.37	0.09
3	2	YY/WW	4.235	1.585	0.61	0.37	0.09
4	1	XXXXXX.00	0.555	0.775	0.61	0.37	0.09
4	2	TTT (optional)	5.155	0.775	0.61	0.37	0.09

NOTE

At Line 3, Step 1, the "RP2-B2" marking denotes device name "RP2" and silicon revision "B2."

5.2. Storage conditions

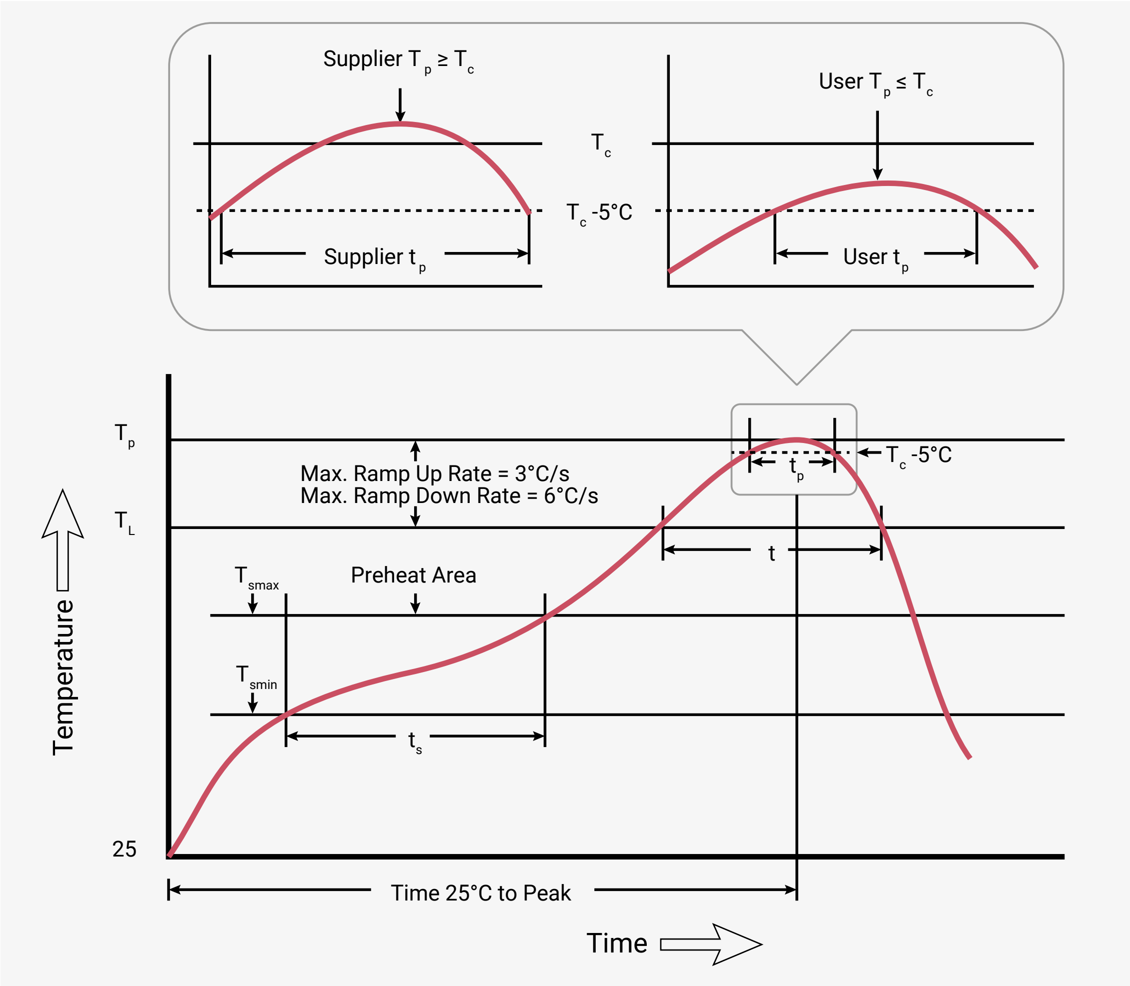
In order to preserve the shelf and floor life of bare RP2040 devices, the recommended storage conditions in line with J-STD (020E & 033D) for RP2040 (classified MSL1) should be kept under 30°C and 85% relative humidity.

5.3. Solder profile

RP2040 is a Pb-free part, with a T_p value of 260°C.

All temperatures refer to the center of the package, measured on the package body surface that is facing up during assembly reflow (live-bug orientation). If parts are reflowed in other than the normal live-bug assembly reflow orientation (i.e., dead-bug), T_p shall be within ±2°C of the live-bug T_p and still meet the T_c requirements; otherwise, the profile shall be adjusted to achieve the latter.

Figure 169. Classification profile (not to scale)



NOTE

Reflow profiles in this document are for classification/preconditioning, and are not meant to specify board assembly profiles. Actual board assembly profiles should be developed based on specific process needs and board designs, and should not exceed the parameters in Table 613.

Table 613. Solder profile values

Profile feature	Value
Temperature min (T_{smin})	150°C
Temperature max (T_{smax})	200°C
Time (t_s) from (T_{smin} to T_{smax})	60 — 120 seconds
Ramp-up rate (T_L to T_p)	3°C/second max.
Liquidous temperature (T_L)	217°C
Time (t_L) maintained above T_L	60 to 150 seconds
Peak package body temperature (T_p)	260°C
Classification temperature (T_c)	260°C
Time (t_p) within 5°C of the specified classification temperature (T_c)	30 seconds
Ramp-down rate (T_p to T_L)	6°C/second max.
Time 25°C to peak temperature	8 minutes max.

5.4. Compliance

RP2040 is compliant to Moisture Sensitivity Level 1.

RP2040 is compliant to the requirement of REACH Substances of Very High Concern (SVHC) that ECHA announced on 25 June 2020.

RP2040 is compliant to the requirement and standard of Controlled Environment-related Substance of RoHS directive (EU) 2011/65/EU and directive (EU) 2015/863.

Package Level reliability qualifications carried out on RP2040:

- Temperature Cycling per JESD22-A104
- HAST per JESD22-A110
- HTSL per JESD22-A103

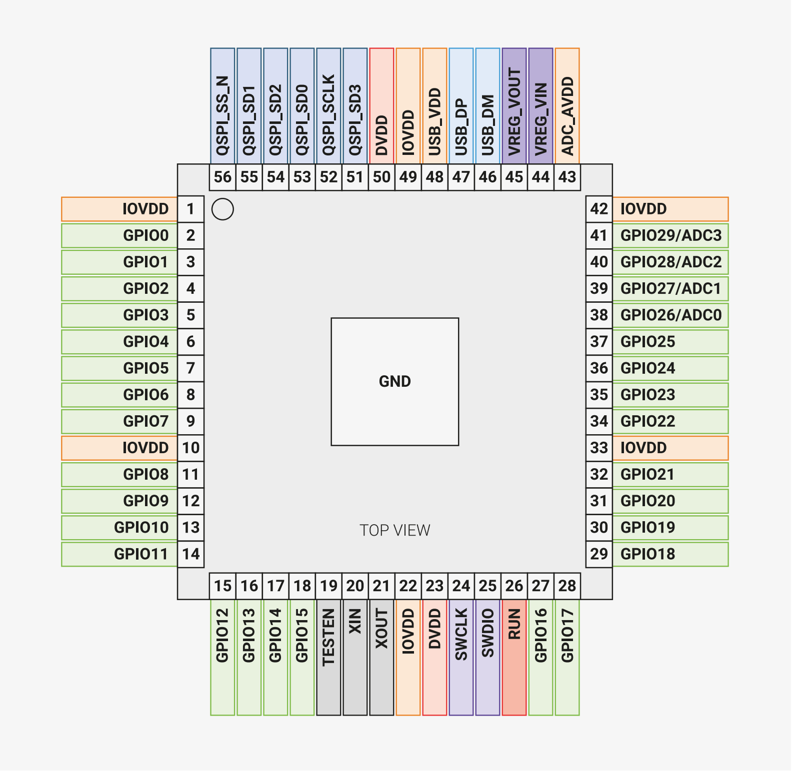
NOTE

A tin whiskers test is not performed as RP2040 is a bottom only termination device (QFN package) which not applicable to JEDEC standard (JESD201A).

5.5. Pinout

5.5.1. Pin Locations

Figure 170. RP2040 QFN-56 package pinout



5.5.2. Pin Definitions

5.5.2.1. Pin Types

In the following GPIO Pin table (Table 615), the pin types are defined as shown below.

Table 614. Pin Types

Pin Type	Direction	Description
Digital In	Input only	**Standard Digital**. Programmable Pull-Up, Pull-Down, Slew Rate, Schmitt Trigger and Drive Strength. Default Drive Strength is 4mA.
Digital IO	Bi-directional	
Digital In (FT)	Input only	**Fault Tolerant Digital**. These pins are described as Fault Tolerant, which in this case means that very little current flows into the pin whilst it is below 3.63V and IOVDD is 0V. There is also enhanced ESD protection on these pins. Programmable Pull-Up, Pull-Down, Slew Rate, Schmitt Trigger and Drive Strength. Default Drive Strength is 4mA.
Digital IO (FT)	Bi-directional	
Digital IO / Analogue	Bi-directional (digital), Input (Analogue)	**Standard Digital and ADC input**. Programmable Pull-Up, Pull-Down, Slew Rate, Schmitt Trigger and Drive Strength. Default Drive Strength is 4mA.
USB IO	Bi-directional	These pins are for USB use, and contain internal pull-up and pull-down resistors, as per the USB specification. **Note** that external 27Ω series resistors are required for USB operation.
Analogue (XOSC)		Oscillator input pins for attaching a 12MHz crystal. Alternatively, XIN may be driven by a square wave.

5.5.2.2. Pin List

Table 615. GPIO pins

Name	Number	Type	Power Domain	Reset State	Description
GPIO0	2	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO1	3	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO2	4	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO3	5	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO4	6	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO5	7	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO6	8	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO7	9	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO8	11	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO9	12	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO10	13	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO11	14	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO12	15	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO13	16	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO14	17	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO15	18	Digital IO (FT)	IOVDD	Pull-Down	User IO

Name	Number	Type	Power Domain	Reset State	Description
GPIO16	27	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO17	28	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO18	29	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO19	30	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO20	31	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO21	32	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO22	34	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO23	35	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO24	36	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO25	37	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO26 / ADC0	38	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input
GPIO27 / ADC1	39	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input
GPIO28 / ADC2	40	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input
GPIO29 / ADC3	41	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input

Table 616. QSPI pins

Name	Number	Type	Power Domain	Reset State	Description
QSPI_SD3	51	Digital IO	IOVDD		QSPI data
QSPI_SCLK	52	Digital IO	IOVDD	Pull-Down	QSPI clock
QSPI_SD0	53	Digital IO	IOVDD		QSPI data
QSPI_SD2	54	Digital IO	IOVDD		QSPI data
QSPI_SD1	55	Digital IO	IOVDD		QSPI data
QSPI_CSn	56	Digital IO	IOVDD	Pull-Up	QSPI chip select

Table 617. Crystal oscillator pins

Name	Number	Type	Power Domain	Description
XIN	20	Analogue (XOSC)	IOVDD	Crystal oscillator. XIN may also be driven by a square wave.
XOUT	21	Analogue (XOSC)	IOVDD	Crystal oscillator.

Table 618. Serial wire debug pins

Name	Number	Type	Power Domain	Reset State	Description
SWCLK	24	Digital In (FT)	IOVDD	Pull-Up	Debug clock
SWD	25	Digital IO (FT)	IOVDD	Pull-Up	Debug data

Table 619. Miscellaneous pins

Name	Number	Type	Power Domain	Reset State	Description
RUN	26	Digital In (FT)	IOVDD	Pull-Up	Chip enable / reset

Name	Number	Type	Power Domain	Reset State	Description
TESTEN	19	Digital In	IOVDD	Pull-Down	Test enable (connect to Gnd)

Table 620. USB pins

Name	Number	Type	Power Domain	Description
USB_DP	47	USB IO	USB_VDD	USB Data +ve. 27Ω series resistor required for USB operation
USB_DM	46	USB IO	USB_VDD	USB Data -ve. 27Ω series resistor required for USB operation

Table 621. Power supply pins

Name	Number(s)	Description
IOVDD	1, 10, 22, 33, 42, 49	IO supply
DVDD	23, 50	Core supply
VREG_VIN	44	Voltage regulator input supply
VREG_VOUT	45	Voltage regulator output
USB_VDD	48	USB supply
ADC_AVDD	43	ADC supply
GND	57	Common ground connection via central pad

5.5.3. Pin Specifications

The following electrical specifications are obtained from characterisation over the specified temperature and voltage ranges, as well as process variation, unless the specification is marked as 'Simulated'. In this case, the data is for information purposes only, and is not guaranteed.

5.5.3.1. Absolute Maximum Ratings

Table 622. Absolute maximum ratings for digital IO (Standard and Fault Tolerant)

Parameter	Symbol	Minimum	Maximum	Units	Comment
I/O Supply Voltage	IOVDD	-0.5	3.63	V	
Voltage at IO	V_{PIN}	-0.5	IOVDD + 0.5	V	

5.5.3.2. ESD Performance

Table 623. ESD performance for all pins, unless otherwise stated

Parameter	Symbol	Maximum	Units	Comment
Human Body Model	HBM	2	kV	Compliant with JEDEC specification JS-001-2012 (April 2012)

Parameter	Symbol	Maximum	Units	Comment
Human Body Model ***Digital (FT) pins only***	HBM	4	kV	Compliant with JEDEC specification JS-001-2012 (April 2012)
Charged Device Model	CDM	500	V	Compliant with JESD22-C101E (December 2009)

5.5.3.3. Thermal Performance

Table 624. Thermal Performance

Parameter	Symbol	Minimum	Typical	Maximum	Units	Comment
Case Temperature	T_C	-40		85	°C	

5.5.3.4. IO Electrical Characteristics

Table 625. Digital IO characteristics - Standard and FT unless otherwise stated

Parameter	Symbol	Minimum	Maximum	Units	Comment
Pin Input Leakage Current	I_{IN}		1	μA	
Input Voltage High @ IOVDD=1.8V	V_{IH}	0.65 * IOVDD	IOVDD + 0.3	V	
Input Voltage High @ IOVDD=2.5V	V_{IH}	1.7	IOVDD + 0.3	V	
Input Voltage High @ IOVDD=3.3V	V_{IH}	2	IOVDD + 0.3	V	
Input Voltage Low @ IOVDD=1.8V	V_{IL}	-0.3	0.35 * IOVDD	V	
Input Voltage Low @ IOVDD=2.5V	V_{IL}	-0.3	0.7	V	
Input Voltage Low @ IOVDD=3.3V	V_{IL}	-0.3	0.8	V	
Input Hysteresis Voltage @ IOVDD=1.8V	V_{HYS}	0.1 * IOVDD		V	Schmitt Trigger enabled
Input Hysteresis Voltage @ IOVDD=2.5V	V_{HYS}	0.2		V	Schmitt Trigger enabled
Input Hysteresis Voltage @ IOVDD=3.3V	V_{HYS}	0.2		V	Schmitt Trigger enabled
Output Voltage High @ IOVDD=1.8V	V_{OH}	1.24	IOVDD	V	I_{OH} = 2, 4, 8 or 12mA depending on setting

Parameter	Symbol	Minimum	Maximum	Units	Comment
Output Voltage High @ IOVDD=2.5V	V_{OH}	1.78	IOVDD	V	I_{OH} = 2, 4, 8 or 12mA depending on setting
Output Voltage High @ IOVDD=3.3V	V_{OH}	2.62	IOVDD	V	I_{OH} = 2, 4, 8 or 12mA depending on setting
Output Voltage Low @ IOVDD=1.8V	V_{OL}	0	0.3	V	I_{OL} = 2, 4, 8 or 12mA depending on setting
Output Voltage Low @ IOVDD=2.5V	V_{OL}	0	0.4	V	I_{OL} = 2, 4, 8 or 12mA depending on setting
Output Voltage Low @ IOVDD=3.3V	V_{OL}	0	0.5	V	I_{OL} = 2, 4, 8 or 12mA depending on setting
Pull-Up Resistance	R_{PU}	50	80	kΩ	
Pull-Down Resistance	R_{PD}	50	80	kΩ	
Maximum Total IOVDD current	I_{IOVDD_MAX}		50	mA	Sum of all current being sourced by GPIO and QSPI pins
Maximum Total VSS current due to IO (IOVSS)	I_{IOVSS_MAX}		50	mA	Sum of all current being sunk into GPIO and QSPI pins

Table 626. USB IO characteristics

Parameter	Symbol	Minimum	Maximum	Units	Comment
Pin Input Leakage Current	I_{IN}		1	μA	
Single Ended Input Voltage High	V_{IHSE}	2		V	
Single Ended Input Voltage Low	V_{ILSE}		0.8	V	
Differential Input Voltage High	V_{IHDIFF}	0.2		V	
Differential Input Voltage Low	V_{ILDIFF}		-0.2	V	
Output Voltage High	V_{OH}	2.8	USB_VDD	V	
Output Voltage Low	V_{OL}	0	0.3	V	
Pull-Up Resistance - RPU2	R_{PU2}	0.873	1.548	kΩ	

Parameter	Symbol	Minimum	Maximum	Units	Comment
Pull-Up Resistance - RPU1&2	$R_{PU1\&2}$	1.398	3.063	kΩ	
Pull-Down Resistance	R_{PD}	14.25	15.75	kΩ	

Table 627. ADC characteristics

Parameter	Symbol	Minimum	Maximum	Units	Comment
ADC Input Voltage Range	V_{PIN_ADC}	0	ADC_AVDD	V	
Effective Number of Bits	ENOB	8.7		bits	See Section 4.9.3
Resolved Bits			12	bits	
ADC Input Impedance	R_{IN_ADC}	100		kΩ	

Table 628. Oscillator pin characteristics when using a Square Wave input

Parameter	Symbol	Minimum	Maximum	Units	Comment
Input Voltage High	V_{IH}	0.65*IOVDD	IOVDD + 0.3	V	XIN only. XOUT floating
Input Voltage Low	V_{IL}	0	0.35*IOVDD	V	XIN only. XOUT floating

See Section 2.16 for more details on the Oscillator, and Minimal Design Example for information on crystal usage.

5.5.3.5. Interpreting GPIO output voltage specifications

The GPIOs on RP2040 have four different output drive strengths, which are nominally called 2, 4, 8 and 12mA modes. These are not hard limits, nor do they mean that they will always be sourcing (or sinking) the selected amount of milliamps. The amount of current a GPIO sources or sinks is dependent on the load attached to it. It will attempt to drive the output to the IOVDD level (or 0V in the case of a logic 0), but the amount of current it is able to source is limited, which will be dependent on the selected drive strength. Therefore the higher the current load is, the lower the voltage will be at the pin. At some point, the GPIO will be sourcing so much current, that the voltage is so low, it won't be recognised as a logic 1 by the input of a connected device. The purpose of the output specifications in Table 625 are to try and quantify how much lower the voltage can be expected to be, when drawing specified amounts of current from the pin.

The Output High Voltage (V_{OH}) is defined as the lowest voltage the output pin can be when driven to a logic 1 with a particular selected drive strength; e.g., 4mA being sourced by the pin whilst in 4mA drive strength mode. The Output Low Voltage is similar, but with a logic 0 being driven.

In addition to this, the sum of all the IO currents being sourced (i.e. when outputs are being driven high) from the IOVDD bank (essentially the GPIO and QSPI pins), must not exceed I_{IOVDD_MAX}. Similarly, the sum of all the IO currents being sunk (i.e. when the outputs are being driven low) must not exceed I_{IOVSS_MAX}.

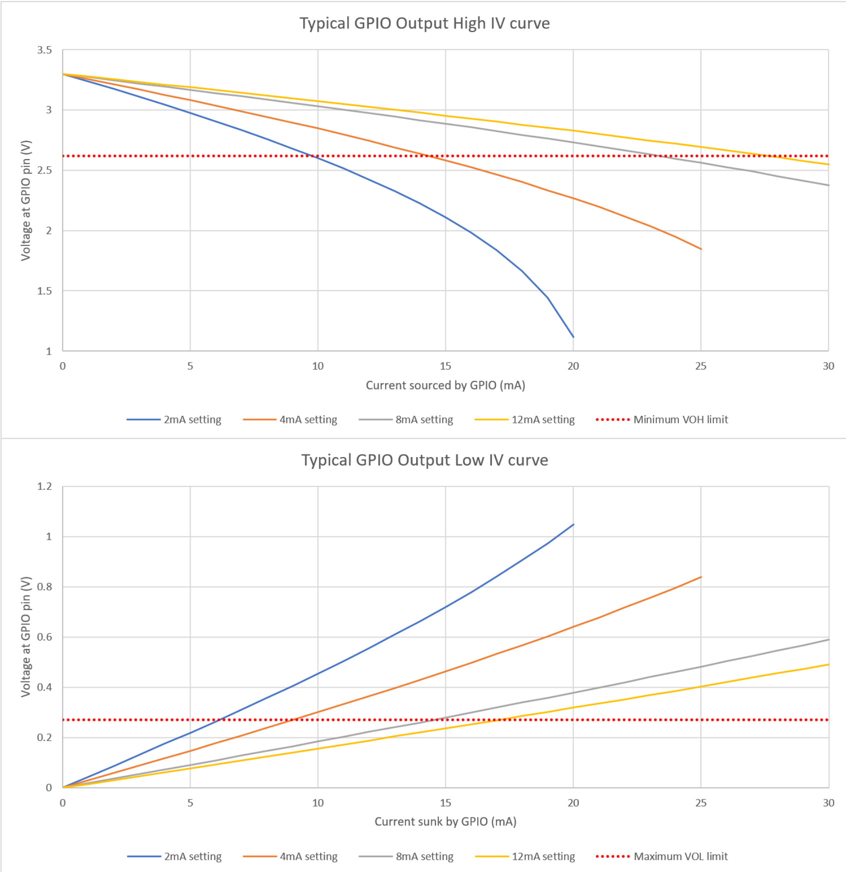


Figure 171. Typical Current vs Voltage curves of a GPIO output.

Figure 171 shows the effect on the output voltage as the current load on the pin increases. You can clearly see the effect of the different drive strengths; the higher the drive strength, the closer the output voltage is to IOVDD (or 0V) for a given current. The minimum V_{OH} and maximum V_{OL} limits are shown in red. You can see that at the specified current for each drive strength, the voltage is well within the allowed limits, meaning that this particular device could drive a lot more current and still be within V_{OH}/V_{OL} specification. This is a typical part at room temperature, there will be a spread of other devices which will have voltages much closer to this limit. Of course, if your application doesn't need such tightly controlled voltages, then you can source or sink more current from the GPIO than the selected drive strength setting, but experimentation will be required to determine if it indeed safe to do so in your application, as it will be outside the scope of this specification.

5.5.3.6. Pin IO Delays

These delays include PIO's input/output mapping logic, IO muxing, and the actual pad delays into a nominal load of 5 pF. Min/max is over the extremes of process variation, voltage (1.1 V +- 10%) and temperature (-40 C to 125 C).

These delays assume an IOVDD of 1.8 V, with `PADS_VSEL` set. At IOVDD = 3.3 V, the delay is significantly lower, and the range is smaller.

The flops themselves have a typical setup time of 10.6 ps and hold time of 2.2 ps. The IO delays between flops and pads must be taken into account.

For minimum and maximum output delays, from `CLK_SYS` arriving at any flop in PIO to the data being valid at a particular GPIO pad see Table 629.

Table 629. Pin minimum and maximum delays from flop to pad, in nanoseconds.

Pad output	Min delay (ns)	Max delay (ns)
GPIO0	2.27	7.10
GPIO1	2.31	7.07

Pad output	Min delay (ns)	Max delay (ns)
GPIO2	2.33	7.08
GPIO3	2.24	7.00
GPIO4	2.30	7.07
GPIO5	2.34	7.10
GPIO6	2.32	7.10
GPIO7	2.39	7.09
GPIO8	2.34	7.09
GPIO9	2.38	7.08
GPIO10	2.33	7.07
GPIO11	2.36	7.08
GPIO12	2.35	7.04
GPIO13	2.31	7.08
GPIO14	2.38	7.06
GPIO15	2.33	7.05
GPIO16	2.34	7.09
GPIO17	2.37	7.09
GPIO18	2.37	7.04
GPIO19	2.27	7.10
GPIO20	2.38	7.09
GPIO21	2.05	7.10
GPIO22	2.34	7.07
GPIO23	2.16	7.05
GPIO24	2.12	7.06
GPIO25	2.26	7.10
GPIO26	2.32	7.09
GPIO27	2.26	7.08
GPIO28	2.34	7.09
GPIO29	2.30	7.07

For minimum and maximum input delays, from pad input to the input synchroniser, see Table 630.

Table 630. Pin minimum and maximum delays from pad input to input synchroniser, in nanoseconds.

Pad output	Min delay (ns)	Max delay (ns)
GPIO1	1.89	5.22
GPIO2	1.84	5.25
GPIO3	1.83	5.24
GPIO4	1.90	5.17
GPIO5	1.90	5.14

Pad output	Min delay (ns)	Max delay (ns)
GPIO6	1.91	5.19
GPIO7	1.91	5.14
GPIO8	1.95	5.14
GPIO9	1.96	5.12
GPIO10	1.95	5.11
GPIO11	1.92	5.16
GPIO12	1.92	5.15
GPIO13	1.94	5.16
GPIO14	1.90	5.18
GPIO15	1.92	5.15
GPIO16	1.95	5.13
GPIO17	1.95	5.12
GPIO18	1.95	5.10
GPIO19	1.95	5.12
GPIO21	2.07	4.98
GPIO23	1.98	5.06
GPIO24	1.97	5.07
GPIO25	1.97	5.08
GPIO26	1.96	5.12
GPIO27	1.94	5.13
GPIO28	1.95	5.13
GPIO29	1.99	5.10

For minimum and maximum input delays over all corners, from pad input to state machine IN data flops (synchronisers bypassed) see Table 631.

Table 631. Pin minimum and maximum delays from pad input to state machine IN data flops (synchronisers bypassed), in nanoseconds.

Pad input	Min delay (ns)	Max delay (ns)
GPIO1	2.22	5.45
GPIO2	2.25	5.49
GPIO3	2.23	5.18
GPIO4	2.24	5.41
GPIO5	2.30	5.65
GPIO6	2.25	5.48
GPIO7	2.26	5.50
GPIO8	2.30	5.51
GPIO9	2.25	5.68
GPIO10	2.34	5.71
GPIO11	2.28	5.47

Pad input	Min delay (ns)	Max delay (ns)
GPIO12	2.29	5.40
GPIO13	2.25	5.47
GPIO14	2.24	5.41
GPIO15	2.23	5.47
GPIO16	2.30	5.42
GPIO17	2.28	5.44
GPIO18	2.28	5.34
GPIO19	2.30	5.50
GPIO21	2.16	5.79
GPIO23	2.33	5.53
GPIO24	2.28	5.60
GPIO25	2.29	5.53
GPIO26	2.28	5.38
GPIO27	2.27	5.39
GPIO28	2.24	5.28
GPIO29	2.33	5.47

5.5.3.6.1. Effects of IOVDD

All of the IO delays given above assume IOVDD = 1.8V. Increasing IOVDD to 3.3V reduces the pad delays quite significantly, and the pad delay is a large fraction of the delays reported above. See Table 632 for a summary of best and worst pad delays at 1.8V and 3.3V.

Table 632. Best and worst pad delays at 1.8V and 3.3V.

Path type	IOVDD	Min delay (ns)	Max delay (ns)
Output	1.8V	1.54	3.65
Output	3.3V	1.11	2.14
Input	1.8V	0.63	1.06
Input	3.3V	0.47	0.76

Changing IOVDD does not affect any logic in the core domain, so these differences can be added to the IO delay tables above to estimate the IO delay ranges at IOVDD = 3.3V (see Table 633).

Table 633. IO delay ranges at IOVDD = 3.3V.

Path group	IOVDD	Min delay (ns)	Max delay (ns)
Output	1.8V	2.12	7.10
Output	3.3V	1.69	5.59
Input to sync	1.8V	1.83	5.25
Input to sync	3.3V	1.67	4.95
Input to SM	1.8V	2.16	5.79
Input to SM	3.3V	2.00	5.49